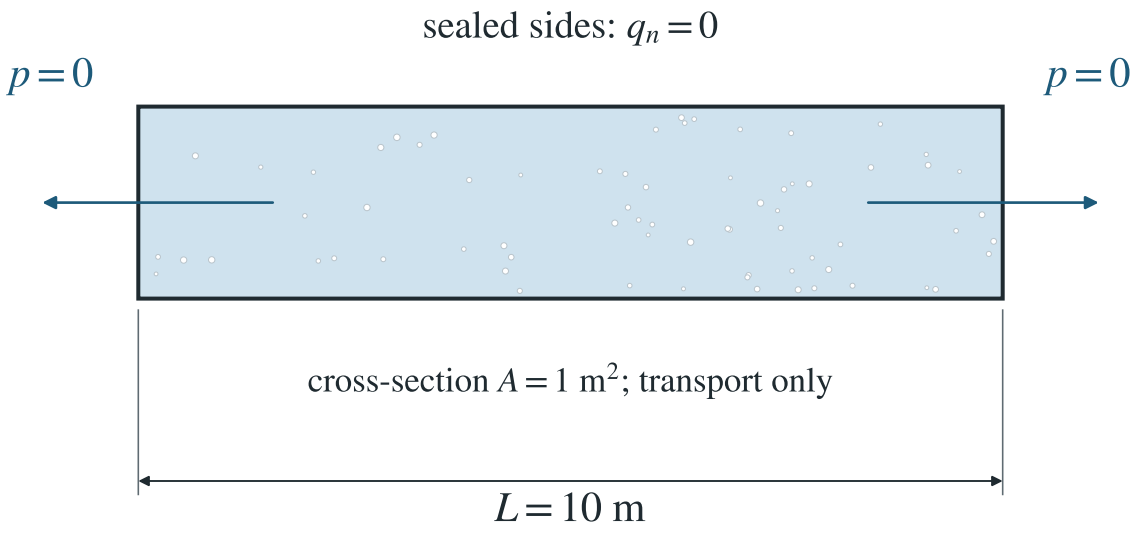$$p(x,0) = P\sin(\pi x/L)$$